

Estimation Theory

Sequential LMMSE Estimator

Sequential LMMSE Estimator

$$x[n] = A + w[n], \quad n = 0, 1, \dots, N-1, \quad \text{where } A \sim \mathcal{N}(\mu_A, \sigma_A^2), w[n] \sim \mathcal{N}(0, \sigma^2)$$

$$\text{MMSE estimator is: } \hat{A} = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}} \bar{x} + \frac{\frac{\sigma^2}{N}}{\sigma_A^2 + \frac{\sigma^2}{N}} \mu_A \quad \text{Bmse}(\hat{A}) = \frac{\sigma^2}{N} \left(\frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}} \right)$$

$$x[n] = A + w[n], \quad n = 0, 1, \dots, N-1 \quad \text{No assumption on distribution of } A \text{ \& } w$$

LMMSE estimator is the same as the MMSE estimator (with the multivariate Gaussian assumption)

Assume $\mu_A = 0$, the LMMSE estimator is:

$$\hat{A}[N-1] = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}} \bar{x} \quad \text{Bmse}(\hat{A}[N-1]) = \frac{\sigma_A^2 \sigma^2}{N \sigma_A^2 + \sigma^2}$$

Sequential LMMSE estimator: Find the LMMSE estimator $\hat{A}[N]$ from $\hat{A}[N-1]$ and $x[N]$ for each N .

new data

Prev. estim.

Sequential LMMSE Estimator

$$x[n] = A + w[n], \quad n = 0, 1, \dots, N-1$$

Assume $\mu_A = 0$, the LMMSE estimator is: $\hat{A}[N-1] = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}} \bar{x}$ $\text{Bmse}(\hat{A}[N-1]) = \frac{\sigma_A^2 \sigma^2}{N\sigma_A^2 + \sigma^2}$

$$\begin{aligned} \hat{A}[N] &= \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N+1}} \frac{1}{N+1} \sum_{n=0}^N x[n] \\ &= \frac{N\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} \frac{1}{N} \left(\sum_{n=0}^{N-1} x[n] + x[N] \right) \\ &= \frac{N\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} \underbrace{\frac{\sigma_A^2 + \frac{\sigma^2}{N}}{\sigma_A^2} \hat{A}[N-1]}_{\text{prod.}} + \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} x[N] \\ &= \frac{N\sigma_A^2 + \sigma^2}{(N+1)\sigma_A^2 + \sigma^2} \hat{A}[N-1] + \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} x[N] \\ &= \hat{A}[N-1] + \left(\frac{N\sigma_A^2 + \sigma^2}{(N+1)\sigma_A^2 + \sigma^2} - 1 \right) \hat{A}[N-1] + \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} x[N] \\ &= \hat{A}[N-1] + \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} (x[N] - \hat{A}[N-1]). \end{aligned} \quad \text{Bmse}(\hat{A}[N]) = \frac{\sigma_A^2 \sigma^2}{(N+1)\sigma_A^2 + \sigma^2}$$

$= \frac{N\sigma_A^2 + \sigma^2}{(N+1)\sigma_A^2 + \sigma^2} \frac{\sigma_A^2 \sigma^2}{N\sigma_A^2 + \sigma^2}$
 $= (1 - K[N]) \text{Bmse}(\hat{A}[N-1])$

$$\begin{aligned} K[N] &= \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} \\ &= \frac{\text{Bmse}(\hat{A}[N-1])}{\text{Bmse}(\hat{A}[N-1]) + \sigma^2} \end{aligned}$$

; gain or scaling factor

Estimator Update:

$$\hat{A}[N] = \hat{A}[N-1] + K[N](x[N] - \hat{A}[N-1])$$

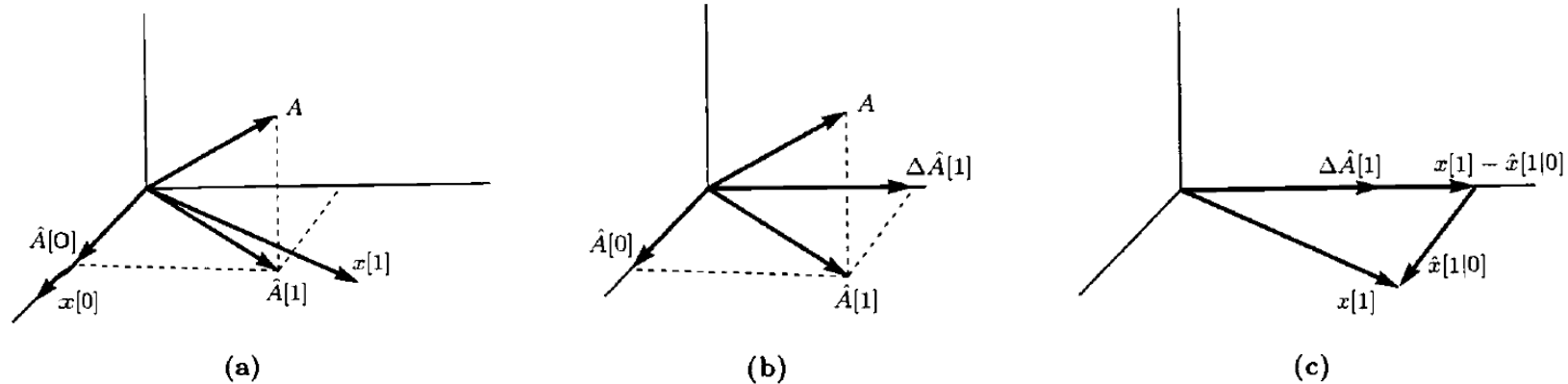
where

$$K[N] = \frac{\text{Bmse}(\hat{A}[N-1])}{\text{Bmse}(\hat{A}[N-1]) + \sigma^2}.$$

Minimum MSE Update:

$$\text{Bmse}(\hat{A}[N]) = (1 - K[N]) \text{Bmse}(\hat{A}[N-1]).$$

Sequential LMMSE Estimator: Geometrical Interpretation



- $\hat{A}[1]$ is the orthogonal projection of A onto the subspace spanned by $x[0]$ and $x[1]$. $\hat{A}[1] = A \times [0] + B \times [1]$
- $\hat{A}[1] = \hat{A}[0] + \Delta \hat{A}[1]$, where $\Delta \hat{A}[1]$ is a component orthogonal to $\hat{A}[0]$.
- Let $\hat{x}[1|0]$ be the LMMSE estimator of $x[1]$ given $x[0]$.
- The error $\tilde{x}[1] = x[1] - \hat{x}[1|0]$ is orthogonal to $x[0]$ (also known as *innovation*).
- $\Delta \hat{A}[1]$ is a projection of A onto $\tilde{x}[1] = x[1] - \hat{x}[1|0]$. : Error Vector.

$$\begin{aligned} \Delta \hat{A}[1] &= \left(A, \frac{\tilde{x}[1]}{\|\tilde{x}[1]\|} \right) \frac{\tilde{x}[1]}{\|\tilde{x}[1]\|} \\ &= \frac{E(A\tilde{x}[1])\tilde{x}[1]}{E(\tilde{x}^2[1])}. \end{aligned}$$

$$K[1] = E(A\tilde{x}[1])/E(\tilde{x}^2[1])$$

$$\hat{A}[1] = \hat{A}[0] + K[1](x[1] - \hat{x}[1|0])$$

$$\hat{A}[1] = \hat{A}[0] + \Delta \hat{A}[1]$$

More generally,

$$\hat{A}[N-1] = \sum_{n=0}^{N-1} K[n](x[n] - \hat{x}[n|0, 1, \dots, n-1])$$

$$K[n] = \frac{E[A(x[n] - \hat{x}[n|0, 1, \dots, n-1])]}{E[(x[n] - \hat{x}[n|0, 1, \dots, n-1])^2]}$$

$$\hat{A}[N] = \hat{A}[N-1] + K[N](x[N] - \hat{x}[N|0, 1, \dots, N-1])$$

Sequential LMMSE Estimator: Derivation

$$\hat{\theta}_i[n] = \hat{\theta}_i[n-1] + K_i[n] (x[n] - \hat{x}[n|n-1])$$

$$\hat{\theta}[n] = \hat{\theta}[n-1] + \mathbf{K}[n] (x[n] - \hat{x}[n|n-1])$$

Before beginning the derivation we state some properties which are needed. The reader should recall that all random variables are assumed to be zero mean.

1. The LMMSE estimator of $\mathbf{A}\theta$ is $\mathbf{A}\hat{\theta}$, the commutative property of (12.23).
2. The LMMSE estimator of $\theta_1 + \theta_2$ is $\hat{\theta}_1 + \hat{\theta}_2$, the additive property of (12.24).
3. Since $\hat{\theta}_i[n-1]$ is a linear combination of the samples $\{x[0], x[1], \dots, x[n-1]\}$, and the innovation $x[n] - \hat{x}[n|n-1]$ is uncorrelated with the past data samples, it follows that

$$E \left[\hat{\theta}_i[n-1] (x[n] - \hat{x}[n|n-1]) \right] = 0.$$

4. Since θ and $w[n]$ are uncorrelated as assumed in the Bayesian linear model and $\hat{\theta}[n-1]$ and $w[n]$ are also uncorrelated, it follows that

$$E \left[\left(\theta - \hat{\theta}[n-1] \right) w[n] \right] = 0.$$

Sequential LMMSE Estimator: Derivation

$$\hat{\boldsymbol{\theta}}[n] = \hat{\boldsymbol{\theta}}[n-1] + \mathbf{K}[n](x[n] - \hat{x}[n|n-1])$$

$$x[n] = \mathbf{h}^T[n]\boldsymbol{\theta} + w[n]$$

(Bayesian linear model - Bayesian G-M theorem)

$$\hat{x}[n|n-1] = \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1] + \hat{w}[n|n-1]$$

(property 1 and 2)

$$\hat{x}[n|n-1] = \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1]$$

(mean zero and property 4)

$$K_i[n] = \frac{E[\theta_i(x[n] - \hat{x}[n|n-1])]}{E[(x[n] - \hat{x}[n|n-1])^2]}$$

$$\begin{aligned} & E[(x[n] - \hat{x}[n|n-1])^2] \\ &= E\left[\left(\mathbf{h}^T[n]\boldsymbol{\theta} + w[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1]\right)^2\right] \\ &= E\left[\left(\mathbf{h}^T[n](\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1]) + w[n]\right)^2\right] \\ &= \mathbf{h}^T[n]E\left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])^T\right]\mathbf{h}[n] \\ &\quad + E(w^2[n]) \\ &= \mathbf{h}^T[n]\mathbf{M}[n-1]\mathbf{h}[n] + \sigma_n^2 \end{aligned}$$

$$K_i[n] = \frac{E[(\theta_i - \hat{\theta}_i[n-1])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])^T]\mathbf{h}[n]}{\mathbf{h}^T[n]\mathbf{M}[n-1]\mathbf{h}[n] + \sigma_n^2}$$

$$\begin{aligned} & E[\theta_i(x[n] - \hat{x}[n|n-1])] \\ &= E[(\theta_i - \hat{\theta}_i[n-1])(x[n] - \hat{x}[n|n-1])] \\ &= E[(\theta_i - \hat{\theta}_i[n-1])(\mathbf{h}^T[n]\boldsymbol{\theta} + w[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1])] \\ &= E[(\theta_i - \hat{\theta}_i[n-1])(\mathbf{h}^T[n](\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1]))] \\ &= E[(\theta_i - \hat{\theta}_i[n-1])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])^T]\mathbf{h}[n] \end{aligned}$$

$$\mathbf{K}[n] = \frac{\mathbf{M}[n-1]\mathbf{h}[n]}{\mathbf{h}^T[n]\mathbf{M}[n-1]\mathbf{h}[n] + \sigma_n^2}$$

Sequential LMMSE Estimator: Derivation

$$\begin{aligned}
 \mathbf{M}[n] &= E \left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n])^T \right] \\
 &= E \left[\left(\cancel{\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1]} - \mathbf{K}[n](x[n] - \hat{x}[n|n-1]) \right) \right. \\
 &\quad \left. \cdot \left(\cancel{\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1]} - \mathbf{K}[n](x[n] - \hat{x}[n|n-1]) \right)^T \right] \\
 &= \mathbf{M}[n-1] - E \left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])(x[n] - \hat{x}[n|n-1]) \mathbf{K}^T[n] \right] \\
 &\quad - \mathbf{K}[n] E \left[(x[n] - \hat{x}[n|n-1])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])^T \right] \\
 &\quad + \mathbf{K}[n] E \left[(x[n] - \hat{x}[n|n-1])^2 \right] \mathbf{K}^T[n].
 \end{aligned}$$

$$\mathbf{K}[n] E \left[(x[n] - \hat{x}[n|n-1])^2 \right] = \mathbf{M}[n-1] \mathbf{h}[n]$$

$$\begin{aligned}
 E \left[(\cancel{\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1]})(x[n] - \hat{x}[n|n-1]) \right] &= E \left[\boldsymbol{\theta}(x[n] - \hat{x}[n|n-1]) \right] \quad (\text{property 3}) \\
 &= \mathbf{M}[n-1] \mathbf{h}[n]
 \end{aligned}$$

$$\begin{aligned}
 \mathbf{M}[n] &= \mathbf{M}[n-1] - \cancel{\mathbf{M}[n-1] \mathbf{h}[n] \mathbf{K}^T[n]} - \mathbf{K}[n] \mathbf{h}^T[n] \mathbf{M}[n-1] \\
 &\quad + \mathbf{M}[n-1] \mathbf{h}[n] \mathbf{K}^T[n] \\
 &= (\mathbf{I} - \mathbf{K}[n] \mathbf{h}^T[n]) \mathbf{M}[n-1].
 \end{aligned}$$

LMMSE estimator for nonzero means (no changes for $\mathbf{M}[n]$ and $\mathbf{K}[n]$)

$$\begin{aligned}
 \hat{\boldsymbol{\theta}}[n] - E(\boldsymbol{\theta}) &= \hat{\boldsymbol{\theta}}[n-1] - E(\boldsymbol{\theta}) + \mathbf{K}[n] \left[x[n] - E(x[n]) - \mathbf{h}^T[n](\hat{\boldsymbol{\theta}}[n-1] - E(\boldsymbol{\theta})) \right] \\
 \hat{\boldsymbol{\theta}}[n] &= \hat{\boldsymbol{\theta}}[n-1] + \mathbf{K}[n] \left[x[n] - \mathbf{h}^T[n] \hat{\boldsymbol{\theta}}[n-1] - (E(x[n]) - \mathbf{h}^T[n] E(\boldsymbol{\theta})) \right]
 \end{aligned}$$

Same as before

$$\begin{aligned}
 E(x[n]) &= E(\mathbf{h}^T[n] \boldsymbol{\theta} + w[n]) \\
 &= \mathbf{h}^T[n] E(\boldsymbol{\theta})
 \end{aligned}$$

Sequential LMMSE Estimator

Estimator Update:

$$\hat{\boldsymbol{\theta}}[n] = \hat{\boldsymbol{\theta}}[n-1] + \mathbf{K}[n](x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1])$$

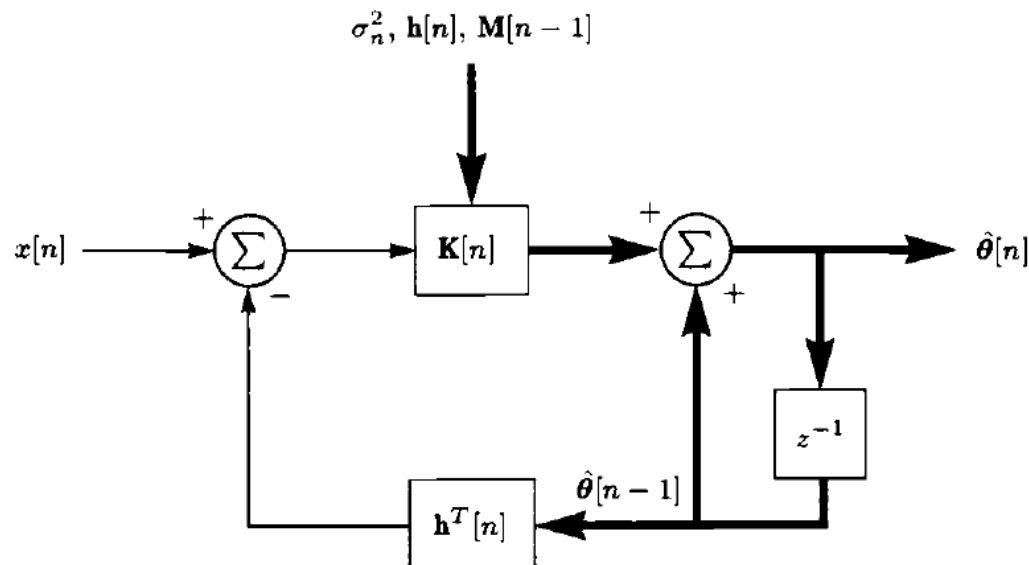
where

$$\mathbf{K}[n] = \frac{\mathbf{M}[n-1]\mathbf{h}[n]}{\sigma_n^2 + \mathbf{h}^T[n]\mathbf{M}[n-1]\mathbf{h}[n]},$$

Minimum MSE Matrix Update:

$$\mathbf{M}[n] = (\mathbf{I} - \mathbf{K}[n]\mathbf{h}^T[n])\mathbf{M}[n-1].$$

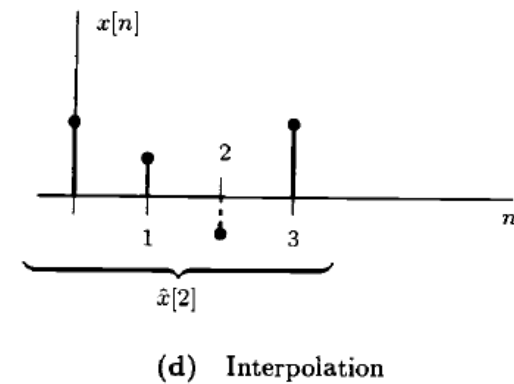
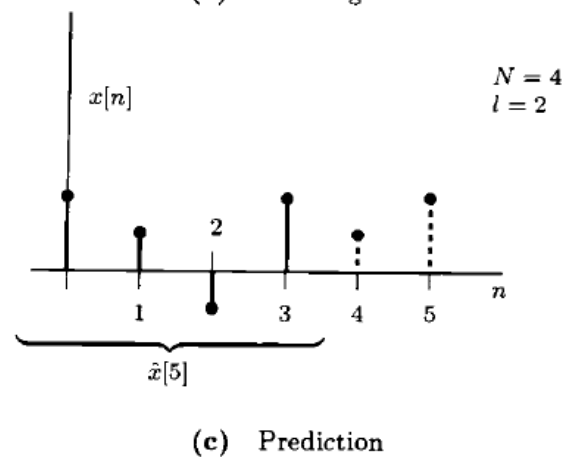
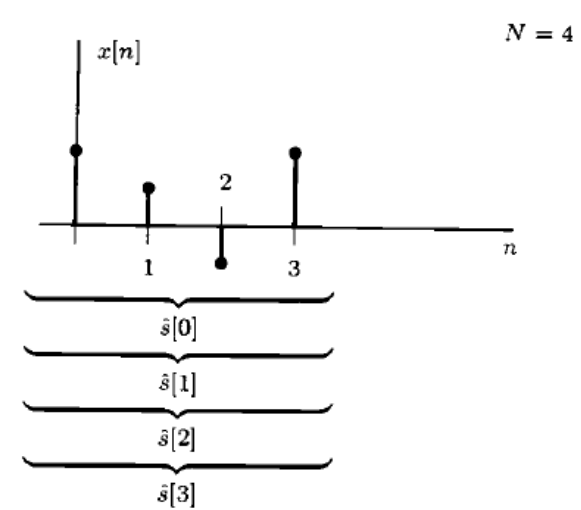
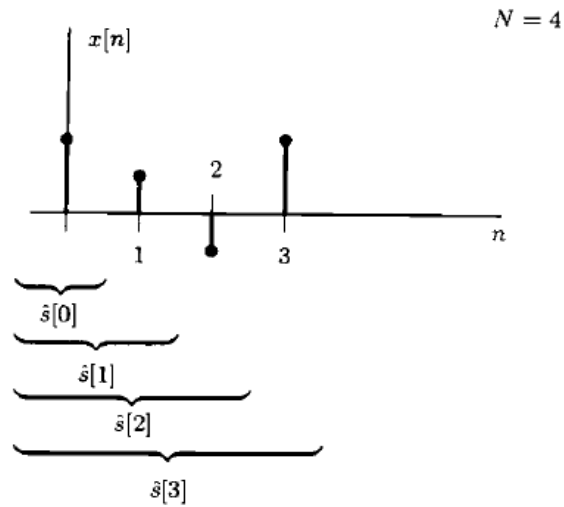
$$\mathbf{M}[n] = E[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n])^T]$$



(cf. Sequential Least Squares)

Estimation Problems

- Filtering
- Smoothing
- Prediction
- Interpolation



Smoothing

$$x[n] = s[n] + w[n]$$

Estimate $\theta = \mathbf{s} = [s[0], s[1], \dots, s[N-1]]^T$ from $\mathbf{x} = [x[0], x[1], \dots, x[N-1]]^T$.

$$\mathbf{r}_{xx}[k] = \mathbf{r}_{ss}[k] + \mathbf{r}_{ww}[k] \quad (\text{signal and noise are uncorrelated})$$

$$\uparrow$$

$$r_{xx}[k] = \mathbb{E}(x[n]x[n+k]), \text{ autocorrelation function}$$

$$\mathbf{C}_{xx} = \mathbf{R}_{xx} = \mathbf{R}_{ss} + \mathbf{R}_{ww}$$

$$\mathbf{C}_{\theta x} = E(\mathbf{s}\mathbf{x}^T) = E(\mathbf{s}(\mathbf{s} + \mathbf{w})^T) = \mathbf{R}_{ss}$$

$$\mathbf{C}_{xx} = \begin{bmatrix} r_{xx}[0] & r_{xx}[1] & \dots & r_{xx}[N-1] \\ r_{xx}[1] & r_{xx}[0] & \dots & r_{xx}[N-2] \\ \vdots & \vdots & \ddots & \vdots \\ r_{xx}[N-1] & r_{xx}[N-2] & \dots & r_{xx}[0] \end{bmatrix}$$

$$= \mathbf{R}_{xx}$$

$$\hat{\mathbf{s}} = \mathbf{R}_{ss}(\mathbf{R}_{ss} + \mathbf{R}_{ww})^{-1}\mathbf{x}$$



Wiener smoothing matrix

$$\hat{\theta} = E(\theta) + \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x}))$$

When $\mathbb{E}(\theta) = 0$ and $\mathbb{E}(\mathbf{x}) = 0$, $\hat{\theta} = \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x}$.

Filtering

$$x[n] = s[n] + w[n]$$

Estimate $\theta = s[n]$ from $\mathbf{x} = [x[0], x[1], \dots, x[n]]^T$.

$$\mathbf{C}_{xx} = \mathbf{R}_{ss} + \mathbf{R}_{ww}$$

$$\begin{aligned} \mathbf{C}_{\theta x} &= E(s[n] [x[0] \ x[1] \ \dots \ x[n]]^T) \\ &= E(s[n] [s[0] \ s[1] \ \dots \ s[n]]^T) \\ &= [r_{ss}[n] \ r_{ss}[n-1] \ \dots \ r_{ss}[0]]^T. \end{aligned}$$

$$\hat{s}[n] = \mathbf{r}_{ss}'^T (\mathbf{R}_{ss} + \mathbf{R}_{ww})^{-1} \mathbf{x}$$

$$\hat{s}[n] = \mathbf{a}^T \mathbf{x} \quad \mathbf{a} = (\mathbf{R}_{ss} + \mathbf{R}_{ww})^{-1} \mathbf{r}_{ss}'$$

$$h^{(n)}[k] = a_{n-k} \quad k = 0, 1, \dots, n.$$

; response of a filter at time n to an impulse applied k samples before.

$$\begin{aligned} \hat{s}[n] &= \sum_{k=0}^n a_k x[k] \\ &= \sum_{k=0}^n h^{(n)}[n-k] x[k] \end{aligned} \quad \hat{s}[n] = \sum_{k=0}^n h^{(n)}[k] x[n-k] \quad ; \text{time-varying FIR filter}$$

$$(\mathbf{R}_{ss} + \mathbf{R}_{ww}) \mathbf{a} = \mathbf{r}_{ss}'$$

Wiener-Hopf filtering equation:

$$(\mathbf{R}_{ss} + \mathbf{R}_{ww}) \mathbf{h} = \mathbf{r}_{ss}$$

$$\begin{bmatrix} r_{xx}[0] & r_{xx}[1] & \dots & r_{xx}[n] \\ r_{xx}[1] & r_{xx}[0] & \dots & r_{xx}[n-1] \\ \vdots & \vdots & \ddots & \vdots \\ r_{xx}[n] & r_{xx}[n-1] & \dots & r_{xx}[0] \end{bmatrix} \begin{bmatrix} h^{(n)}[0] \\ h^{(n)}[1] \\ \vdots \\ h^{(n)}[n] \end{bmatrix} = \begin{bmatrix} r_{ss}[0] \\ r_{ss}[1] \\ \vdots \\ r_{ss}[n] \end{bmatrix}$$